

System and Method for Dynamic Resource Allocation in Distributed Computing Environments

Date of Patent: April 20, 2021

ABSTRACT

A system and method for dynamically allocating computational resources in a distributed computing environment is disclosed. The system comprises a resource manager configured to monitor workload metrics across a plurality of compute nodes, and to redistribute tasks according to a load-balancing algorithm that minimizes tail latency while satisfying quality-of-service constraints. The method includes steps of collecting real-time telemetry, predicting future resource demand using a regression model, and preemptively migrating workloads to underutilized nodes prior to resource exhaustion.

TECHNICAL FIELD The present disclosure relates to distributed computing, and more particularly to systems and methods for dynamic allocation of computational resources across a plurality of networked compute nodes.

BACKGROUND Distributed computing environments face the challenge of efficiently allocating finite computational resources—CPU cycles, memory, and network bandwidth—across workloads with variable and unpredictable demand profiles. Existing load balancers operate reactively, redistributing workloads only after resource exhaustion is detected, resulting in degraded performance and service interruptions. **SUMMARY OF THE INVENTION** In one aspect, the invention provides a proactive resource allocation method that predicts future resource demand and migrates workloads preemptively, reducing tail latency and improving overall system throughput.

CLAIMS

1. A computer-implemented method for dynamic resource allocation in a distributed computing environment, comprising: collecting, by a resource manager executing on a management node, real-time telemetry data from a plurality of compute nodes; predicting future resource demand for each compute node based on the collected telemetry data using a linear regression model; and migrating one or more workloads from a first compute node predicted to exceed a resource threshold to a second compute node predicted to have available capacity, prior to the resource threshold being reached on the first compute node.
2. The method of claim 1, wherein collecting real-time telemetry data comprises sampling CPU utilization, memory utilization, and network throughput metrics at intervals of no greater than one second.
3. The method of claim 1, wherein migrating one or more workloads comprises a live migration protocol that preserves in-flight network connections of the migrated workload.
4. A system comprising: one or more processors; and a non-transitory computer-readable medium storing instructions that, when executed by the one or more processors, implement the method of claim 1.